

BRAF fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes BRAF gene fusions from the msk_impact_50k_2026 study, covering 179 fusion events. 151/179 fusions (84.4%) were in-frame. 163/179 fusions (91.1%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ($p=0.0133676$, statistically significant at $\alpha=0.05$). Compared to the literature benchmark (PMC5461196: 100.0% in-frame, 100.0% domain-retained), this run observed 84.4% in-frame and 91.1% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 179 analyzed fusion events, identifying 67 distinct fusion partner genes. The most frequent partner was KIAA1549, observed in 43 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions retained the domain, $p=0.0133676$ (statistically significant at $\alpha=0.05$). The Protein kinase domain appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.045954.

Domain disruption

Disruption of the RAS-binding domain and Cysteine-rich domain was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions disrupted the domain, $p=0.041705$ (statistically significant at $\alpha=0.05$). The RAS-binding domain and Cysteine-rich domain appear to require loss or disruption rather than retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 1.

Cutpoint detection

Cutpoint detection scanned 178 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 327 aa (permutation-corrected $p=0.019802$, statistically significant at $\alpha=0.05$). This is 47 aa from the nearest configured domain boundary at 280 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=163$) and "not_retained" ($n=15$). For Frame_status, retained: 142/152 (93.4%, 95% CI 88.3%-96.4%); not_retained: 9/11 (81.8%, 95% CI 52.3%-94.9%). Welch's t-test comparing Tumor_variant_count between groups gave means of 52.0736 vs 36.4, $p=0.11965$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (67 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: BRAF has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
AGK	14	0.0782122905027933	0.13376762457977448	0.0	0.9846938775510204	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2792922449674053	1
ABCC2	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	2
ITGB5	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	3
MKLN1	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	4
NEO1	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	5
PWWP2A	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	6
TBXAS1	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	7
UBE2H	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	8
WEE2	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIAA1549	43	0.24022346368715083	0.13376762457977448	0.0	0.436046511627907	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.24559849203424552	10
CLIP1	2	0.0111731843575419	0.13376762457977448	0.0	0.7589285714285714	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22531571720546253	11
LUC7L2	1	0.00558659217877095	0.13376762457977448	0.0	0.6428571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20622902526611697	12
CUL1	5	0.027932960893854747	0.13376762457977448	0.0	0.5249999999999999	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19525436445207067	13
SND1	17	0.09497206703910614	0.13376762457977448	0.0	0.3897058823529411	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19507197864858727	14
FAM131B	3	0.01675977653631285	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19083098057337952	15
CARMIL1	2	0.0111731843575419	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19049428863403395	16
MAD1L1	2	0.0111731843575419	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19049428863403395	17
AKAP9	2	0.0111731843575419	0.13376762457977448	0.0	0.5223214285714286	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18982464577689112	18
EXOC4	2	0.0111731843575419	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18915500291974824	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ABCC1	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	20
CAPZA2	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	21
KLHL12	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	22
LRBA	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	23
OSBPL7	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	24
PRIM2	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	25
PRKAR1B	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	26
PTPRZ1	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	27
RGS3	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	28
TRA2B	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	29

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TRIM24	8	0.0446927374301676	0.13376762457977448	0.0	0.4419642857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18782694027010738	30
CCDC6	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	31
CMAS	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	32
ERC1	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	33
MINDY4	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	34
PHTF2	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	35
SNX8	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	36
AGAP3	6	0.0335195530726257	0.13376762457977448	0.0	0.4196428571428571	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1811267706771305	37
NRF1	2	0.0111731843575419	0.13376762457977448	0.0	0.4642857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18111928863403398	38
GTF2IRD1	3	0.01675977653631285	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.17475955200195098	39

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TMPRSS2	4	0.0223463687150838	0.13376762457977448	0.0	0.37797619047619047	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.17152481536986797	40
ATF7	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	41
CLEC2L	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	42
DBI	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	43
DOCK4	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	44
GIPC2	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	45
GTF2I	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	46
KCND2	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	47
NFIA	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	48
OSBPL9	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	49

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
PAK2	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	50
PRKAR2B	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	51
RRBP1	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	52
SETBP1	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	53
SLC45A3	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	54
SPOCK1	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	55
ZC3HAV1	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	56
ZNF207	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	57
CDK5RAP2	4	0.0223463687150838	0.13376762457977448	0.0	0.3660714285714286	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1697391010841537	58
MKRN1	10	0.055865921787709494	0.13376762457977448	0.0	0.0625	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1342592527202271	59

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
EPB41	1	0.00558659217877095	0.13376762457977448	0.0	0.008928571428571397	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11113973955183125	60
PJA2	1	0.00558659217877095	0.13376762457977448	0.0	0.008928571428571397	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11113973955183125	61
RBM33	1	0.00558659217877095	0.13376762457977448	0.0	0.008928571428571397	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11113973955183125	62
RBMS3	1	0.00558659217877095	0.13376762457977448	0.0	0.008928571428571397	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11113973955183125	63
PLPP1	1	0.00558659217877095	0.13376762457977448	0.0	0.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10980045383754554	64
SCRIB	1	0.00558659217877095	0.13376762457977448	0.0	0.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10980045383754554	65
SVOPL	1	0.00558659217877095	0.13376762457977448	0.0	0.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10980045383754554	66
TRIM33	1	0.00558659217877095	0.13376762457977448	0.0	0.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10980045383754554	67

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
238	1	162	0	15	None	1.0	-0.0
287	1	162	1	14	0.08641975308641975	0.16187392877547135	0.7908230925727642
303	2	161	1	14	0.17391304347826086	0.23330512802756184	0.6320757153442778

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
327	19	144	6	9	0.19791666666666666	0.00883450422715351	2.053817817119616
380	68	95	8	7	0.6263157894736842	0.422537724516002	0.37413451081430776
381	114	49	9	6	1.5510204081632653	0.5596655980047007	0.2520713878510084
393	140	23	12	3	1.5217391304347827	0.46313499579777323	0.33429240123968906
400	141	22	12	3	1.6022727272727273	0.44662922460047316	0.3500528628421283
438	156	7	13	2	3.4285714285714284	0.16897876327149897	0.7721678727445281
439	163	0	14	1	None	0.08426966292134831	1.0743287432532127

domain_disruption tables

frame_domain_contingency_table

142	22
9	5

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
ras_binding	RAS-binding domain	178	14	0	164	0.0
cysteine_rich	Cysteine-rich domain	178	14	1	163	0.005699893955461294

domain_retention tables

frame_domain_contingency_table

142	21
9	6

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	178	163	0	15	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	163	179	0.9106145251396648

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ABCC1	1
ABCC2	1
AGAP3	6
AGK	14
AKAP9	2
ATF7	1
CAPZA2	1
CARMIL1	2
CCDC6	1
CDK5RAP2	4
CLEC2L	1
CLIP1	2
CMAS	1
CUL1	5
DBI	1
DOCK4	1
EPB41	1
ERC1	1
EXOC4	2
FAM131B	3
GIPC2	1
GTF2I	1
GTF2IRD1	3
ITGB5	1
KCND2	1
KIAA1549	43
KLHL12	1
LRBA	1
LUC7L2	1

Partner_gene	Event_count
MAD1L1	2
MINDY4	1
MKLN1	1
MKRN1	10
NEO1	1
NFIA	1
NRF1	2
OSBPL7	1
OSBPL9	1
PAK2	1
PHTF2	1
PJA2	1
PLPP1	1
PRIM2	1
PRKAR1B	1
PRKAR2B	1
PTPRZ1	1
PWWP2A	1
RBM33	1
RBMS3	1
RGS3	1
RRBP1	1
SCRIB	1
SETBP1	1
SLC45A3	1
SND1	17
SNX8	1
SPOCK1	1
SVOPL	1
TBXAS1	1
TMPRSS2	4
TRA2B	1
TRIM24	8
TRIM33	1
UBE2H	1

Partner_gene	Event_count
WEE2	1
ZC3HAV1	1
ZNF207	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 1458-T01-I M6-1	P-0021 458-T0 1-IM6		ABCC 1-BRAF	ABCC C1	in-frame	threonine_prime	140493 152	8	380	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		ABCC1 (NM_004996) - BRAF (NM_004333) rearrangement: t(7;16)(q34, p13.11)(chr7:g.140493152:: chr16:g.16116105) Protein Fusion: in frame {ABCC1:BRAF}	NA
EVT-P-003 4621-T01-I M6-2	P-0034 621-T0 1-IM6		ABCC 2-BRAF	ABCC C2	unknown	threonine_prime	140495 500	8	327	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		ABCC2 (NM_000392) - BRAF (NM_004333) rearrangement: t(7;10)(q34; 24.2)(chr7:g.140495500::chr 10:g.101559038) Protein Fusion: mid-exon {ABCC2:BRAF}	NA
EVT-P-005 2582-T01-I M6-3	P-0052 582-T0 1-IM6		AGAP 3-BRAF	AGAP P3	unknown	threonine_prime	140485 658	9	393	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1637:AGAP3_c.1177+169 0:BRAFInv Protein Fusion: mid-exon {AGAP3:BRAF}	NA
EVT-P-005 0574-T02-I M6-4	P-0050 574-T0 2-IM6		AGAP 3-BRAF	AGAP P3	in-frame	threonine_prime	140493 208	8	380	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1222-2059:AGAP3_c.114 0+900:BRAFInv Protein Fusion: in frame {AGAP3:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 9799-T01-I M6-5	P-0039 799-T01-IM6		AGAP3-BRAF	AGAP3	in-frame	three_prime	140482317	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1221+416:AGAP3_c.1314+504:BRAFInv Protein Fusion: in frame {AGAP3:BRAF}	NA
EVT-P-005 0931-T01-I M6-6	P-0050 931-T01-IM6		AGK-BRAF	AGK	in-frame	three_prime	140499251	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-14294:AGK_c.980+911:BRAFInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-004 2835-T03-I M6-7	P-0042 835-T03-IM6		AGK-BRAF	AGK	in-frame	three_prime	140497995	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-004 2835-T02-I M6-8	P-0042 835-T02-IM6		AGK-BRAF	AGK	in-frame	three_prime	140497995	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-005 0635-T01-I M6-9	P-0050 635-T01-IM6		AKAP9-BRAF	AKAP9	in-frame	three_prime	140487501	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.6613-1489:AKAP9_c.1141-117:BRAFInv Protein Fusion: in frame {AKAP9:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 0237-T03-IM6-10	P-0030 237-T03-IM6		AKAP9-BRAF	AKAP9	in-frame	three_prime	140492357	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.3318+1378:AKAP9_c.1140+1751:BRAF Protein Fusion: in frame {AKAP9:BRAF}	NA
EVT-P-002 3821-T01-IM6-12	P-0023 821-T01-IM6		ATF7-BRAF	ATF7	in-frame	three_prime	140483233	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ATF7 (NM_006856) - BRAF (NM_004333) Rearrangement : t(7;12)(q34;q13.13)(chr7:g.140483233::chr12:g. 53913299) Protein Fusion: in frame {ATF7:BRAF}	NA
EVT-P-000 7861-T01-IM5-13	P-0007 861-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	three_prime	140490697	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1326+220:AGAP3_c.1141-3313:BRAF Protein fusion: in frame (AGAP3-BRAF)	NA
EVT-P-002 1519-T02-IM6-14	P-0021 519-T02-IM6		BRAF-AGAP3	AGAP3	in-frame	three_prime	140493112	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) Rearrangement : c.1326+142:AGAP3_c.1140+996:BRAF Protein Fusion: in frame {AGAP3:BRAF}	NA
EVT-P-000 9757-T01-IM5-15	P-0009 757-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	three_prime	140493871	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1222-2154:AGAP3_c.1140+237:BRAF Protein fusion: in frame (AGAP3-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 6886-T01-I M6-16	P-0036 886-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140494309	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+10859:AGK_c.981-42:BRAFInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-002 5561-T03-I M6-17	P-0025 561-T03-IM6		BRAF-AGK	AGK	in-frame	three_prime	140498719	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 8-18): c.980+1288:BRAF_c.101+772:AGKInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-002 5561-T01-I M6-18	P-0025 561-T01-IM6		BRAF-AGK	AGK	in-frame	five_prime	140498874	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - AGK (NM_018238) Rearrangement: c.980+1288:BRAF_c.101+772:AGKInv Protein Fusion: in frame {BRAF:AGK}	NA
EVT-P-002 4227-T01-I M6-19	P-0024 227-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140495070	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+16796:AGK_c.981-803:BRAFInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-000 4477-T01-I M5-20	P-0004 477-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140495405	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1 to 2 and BRAF exons 8 to 18): c.101+10076:AGK_c.981-1138:BRAFInv Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-002 3378-T01-I M6-21	P-0023 378-T01-IM6		BRAF-AGK	AGK	in-frame	five_prime	140498362	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - AGK (NM_018238) rearrangement: c.980+1800:BRAF_c.101+14574:AGKInv Protein Fusion: in frame {BRAF:AGK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0001832-T02-IM6-22	P-0001832-T02-IM6		BRAF-AGK	AGK	in-frame	three_prime	140494489	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exon 1 fused in-frame to BRAF exons 8-18): c.102-572:AGK_c.981-222:BRAF in v Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-0000913-T02-IM5-23	P-0000913-T02-IM5		BRAF-AGK	AGK	in-frame	three_prime	140496386	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) Fusion (AGK exon 2 fused with BRAF exon 8) : c.101+13932:AGK_c.981-2119:BRAF in v Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-0029531-T01-IM6-24	P-0029531-T01-IM6		BRAF-AGK	AGK	unknown	three_prime	140500233	7	303	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 7-18): c.101+4572:AGK_c.909:BRAF in v Protein Fusion: mid-exon {AGK:BRAF}	NA
EVT-P-0030746-T01-IM6-25	P-0030746-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140497533	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.101+8348:AGK_c.980+2629:BRAF in v Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0010945-T01-IM5-26	P-0010945-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140497610	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+12646:AGK_c.980+2552:BRAF in v Protein fusion: in frame (AGK-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 9035-T01-I M6-85	P-0049 035-T0 1-IM6		BRAF- CLEC2L	CLEC2L	in-frame	five_prime	140485799	9	393	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - CLEC2L (NM_001080511) rearrangement: c.1177+1549:BRAF_c.191-5932:CLEC2Linv Protein Fusion: in frame {BRAF:CLEC2L}	NA
EVT-P-000 4795-T01-I M5-87	P-0004 795-T0 1-IM5		BRAF- CUL1	CUL1	in-frame	three_prime	140493811	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1(NM_003592) - BRAF (NM_004333) Rearrangement :c.790-427: CUL1_c.1140+297:BRAFinv Protein fusion: in frame (CUL1-BRAF)	NA
EVT-P-002 2257-T01-I M6-88	P-0022 257-T0 1-IM6		BRAF- EPB41	EPB41	in-frame	three_prime	140482687	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EPB41 (NM_001166005) - BRAF (NM_004333) fusion (EPB41 exons 1-5 fused in-frame with BRAF exons 11-18): t(1;7)(p35.4;q34)(chr1:g.29424227::chr7:g.140482687) Protein Fusion: in frame {EPB41:BRAF}	NA
EVT-P-000 1988-T01-I M3-89	P-0001 988-T0 1-IM3		BRAF- FAM131B	FAM131B	in-frame	three_prime	140489629	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_001031690) - BRAF (NM_004333) Deletion : c.175-103:FAM131B_c.1141-2245:BRAFdel Protein fusion: in frame (FAM131B-BRAF)	NA
EVT-P-001 0494-T01-I M5-90	P-0010 494-T0 1-IM5		BRAF- GIPC2	GIPC2	in-frame	three_prime	140487276	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655) -BRAF (NM_004333) Rearrangement : t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) Protein fusion: in frame (GIPC2-BRAF)	NA
EVT-P-005 4046-T01-I M6-93	P-0054 046-T0 1-IM6		BRAF- GTF2IRD1	GTF2IRD1	in-frame	five_prime	140484706	10	393	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - GTF2IRD1 (NM_005685) rearrangement: c.1178-1749:BRAF_c.2320+10446:GTF2IRD1inv Protein Fusion: in frame {BRAF:GTF2IRD1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-000 0828-T03-I M6-94	P-0000 828-T0 3-IM6		BRAF- GTF2I RD1	GTF 2IR D1	in-frame	five_prime	140483 554	10	393	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - GTF2IRD1 (NM_005685) rearrangement: c.1178-597: BRAF_c.2320+14181:GTF2IRD1inv Protein Fusion: in frame {BRAF:GTF2IRD1}	NA
EVT-P-001 5847-T02-I M6-95	P-0015 847-T0 2-IM6		BRAF- ITGB5	ITGB5	out-of-frame	five_prime	140497 019	8	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - ITGB5 (NM_002213) Rearrangement : t(3,7)(q21.2;q34)(chr3:g.124579522::chr7:g.140497019) Protein Fusion: out of frame {BRAF:ITGB5}	NA
EVT-P-001 4377-T01-I M6-99	P-0014 377-T0 1-IM6		BRAF- NFIA	NFIA	in-frame	three_prime	140486 673	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NFIA (NM_001145512) - BRAF (NM_004333) rearrangement: t(1;7)(p31.3;q34)(chr1:g.61836194::chr7:g.140486673) Protein fusion: in frame (NFIA-BRAF)	NA
EVT-P-001 0398-T01-I M5-100	P-0010 398-T0 1-IM5		BRAF- OSBPL9	OSBPL9	in-frame	three_prime	140484 912	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		OSBPL9 (NM_148909) - BRAF (NM_004333) Rearrangement : t(1;7)(p32.3;q34)(chr1:g.52228623::chr7:g.140484912) Protein fusion: in frame (OSBPL9-BRAF)	NA
EVT-P-000 3502-T01-I M5-101	P-0003 502-T0 1-IM5		BRAF- PJA2	PJA2	in-frame	five_prime	140482 379	10	438	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA Protein fusion: in frame (BRAF-PJA2)	NA
EVT-P-003 6870-T01-I M6-102	P-0036 870-T0 1-IM6		BRAF- PRIM2	PRIM2	in-frame	three_prime	140493 232	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRIM2 (NM_000947) - BRAF (NM_004333) Rearrangement : t(6;7)(p11.1;q34)(chr6:g.57410815::chr7:g.140493232) Protein Fusion: in frame {PRIM2:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000 9246-T01-I M5-103	P-0009 246-T0 1-IM5		BRAF- RBM3 3	RB M33	in-frame	three_prime	140482 661	10	438	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		RBM33 (NM_053043) - BRAF (NM_004333) rearrangement: c.171+526: RBM33_c.1314+160:BRAF inv Protein fusion: in frame (RBM33-BRAF)	NA
EVT-P-001 6313-T01-I M6-104	P-0016 313-T0 1-IM6		BRAF- SLC4 5A3	SLC 45A 3	unknown	three_prime	140486 608	9	393	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SLC45A3 (NM_033102) - BRAF (NM_004333) fusion (SLC45A3 exons 1 fused to BRAF exons 10-18): t(1;7)(q 32.1;q34)(chr1:g.20563845 2::chr7:g.140486608) Transcript Fusion {SLC45A3:BRAF}	NA
EVT-P-006 4764-T01-I M7-105	P-0064 764-T0 1-IM7		BRAF- SND1	SND 1	in-frame	five_prime	140493 289	8	380	True	lost	0.0	False	RAS-bi nding domain; Cysteine- rich domain	Protein kinase domain		BRAF (NM_004333) - SND1 (NM_014390) fusion: c.1140 +819:BRAF_c.1039-3950:S ND1inv Protein Fusion: in frame {BRAF:SND1}	NA
EVT-P-002 9981-T01-I M6-106	P-0029 981-T0 1-IM6		BRAF- SPOC K1	SPO CK1	in-frame	three_prime	140483 862	10	393	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SPOCK1 (NM_004598) - BRAF (NM_004333) fusion: t(5;7)(q31.2;q34)(chr5:g.136 401840::chr7:g.140483862) Protein Fusion: in frame {SPOCK1:BRAF}	NA
EVT-P-006 4223-T01-I M7-107	P-0064 223-T0 1-IM7		BRAF- TBXA S1	TBX AS1	out-of-frame	five_prime	140494 512	8	327	True	lost	0.0	False	RAS-bi nding domain; Cysteine- rich domain	Protein kinase domain		BRAF (NM_004333) - TBXAS1 (NM_001061) rearrangement: c.981-245:B RAF_c.336+2985:TBXAS1 inv Protein Fusion: out of frame {BRAF:TBXAS1}	NA
EVT-P-002 1469-T01-I M6-108	P-0021 469-T0 1-IM6		BRAF- TRIM3 3	TRI M33	in-frame	three_prime	140481 644	11	439	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		TRIM33 (NM_015906) - BRAF(NM_004333) rearrangement: t(1;7)(p13.2; q34)(chr1:g.114963985::chr 7:g.140481644) Protein Fusion: in frame {TRIM33:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-004 2835-T03-IM6-109	P-0042 835-T03-IM6		BRAF-WEE2	WEE2	unknown	five_prime	140498912	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - WEE2 (NM_001105558) rearrangement: c.980+1250 :BRAF_c.1305:WEE2inv Protein Fusion: mid-exon {BRAF:WEE2}	NA
EVT-P-001 5985-T03-IM6-111	P-0015 985-T03-IM6		CAPZA2-BRAF	CAPZA2	in-frame	three_prime	140491425	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CAPZA2 (NM_006136) - BRAF (NM_004333) fusion: c.40-938:CAPZA2_c.1140+2683:BRAFinv Protein Fusion: in frame {CAPZA2:BRAF}	NA
EVT-P-003 3764-T04-IM6-112	P-0033 764-T04-IM6		CARMIL1-BRAF	CARMIL1	in-frame	three_prime	140491509	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		LRRRC16A (NM_017640) - BRAF (NM_004333) fusion: t(6;7)(p22.2;q34)(chr6:g.25554753::chr7:g.140491509) Protein Fusion: in frame {LRRRC16A:BRAF}	NA
EVT-P-003 0796-T01-IM6-113	P-0030 796-T01-IM6		CARMIL1-BRAF	CARMIL1	in-frame	three_prime	140491645	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		LRRRC16A (NM_017640) - BRAF (NM_004333) fusion (LRRRC16A exons 1-29 fused to BRAF exons 9-18): t(6;7)(p22.2;q34)(chr6:g.25580104::chr7:g.140491645) Protein Fusion: in frame {LRRRC16A:BRAF}	NA
EVT-P-000 5834-T01-IM5-114	P-0005 834-T01-IM5		CCDC6-BRAF	CCDC6	in-frame	three_prime	140488401	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CCDC6 (NM_005436) -BRAF (NM_004333) Rearrangement : t(10;7)(chr10q21.2;q34)(chr10:g.61627617::chr7:g.140488401) Protein fusion: in frame (CCDC6-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 8194-T01-I M6-115	P-0038 194-T0 1-IM6		CDK5 RAP2- BRAF	CDK 5RAP P2	in-frame	three_prime	140484 954	10	393	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement: t(7;9)(q34;q33.2)(chr7:g.140484954::chr9:g.123187156) Protein Fusion: in frame {CDK5RAP2:BRAF}	NA
EVT-P-000 4203-T01-I M5-116	P-0004 203-T0 1-IM5		CDK5 RAP2- BRAF	CDK 5RAP P2	in-frame	three_prime	140491 585	8	380	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement: t(7;9)(7q34;q33.2)(chr7:140491585::chr9:123251876) Protein fusion: in frame (CDK5RAP2-BRAF)	NA
EVT-P-000 5461-T02-I M5-117	P-0005 461-T0 2-IM5		CDK5 RAP2- BRAF	CDK 5RAP P2	in-frame	three_prime	140481 684	11	439	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement : t(7;9)(7q34;q33.2)(chr7:140481684::chr9:123255452) Protein fusion: in frame (CDK5RAP2-BRAF)	NA
EVT-P-001 4969-T01-I M6-118	P-0014 969-T0 1-IM6		CDK5 RAP2- BRAF	CDK 5RAP P2	in-frame	five_prime	140493 700	8	380	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - CDK5RAP2 (NM_018249) Rearrangement : t(7;9)(q34;q33.3)(chr7:g.140493700::chr9:g.123269531) Protein fusion: in frame (BRAF-CDK5RAP2)	NA
EVT-P-004 0461-T01-I M6-119	P-0040 461-T0 1-IM6		CLIP1 -BRAF	CLIP1	in-frame	three_prime	140488 218	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		CLIP1 (NM_001247997) - BRAF (NM_004333) fusion: t(7;12)(q34;q24.31)(chr7:g.140488218::chr12:g.122833264) Protein Fusion: in frame {CLIP1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0007266-T02-IM6-120	P-0007266-T02-IM6		CLIP1-BRAF	CLIP1	in-frame	three_prime	140495673	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CLIP1 (NM_001247997) - BRAF (NM_004333) fusion: t(7;12)(q34;q24.31)(chr7:g.140495673::chr12:g.122767026) Protein Fusion: in frame {CLIP1:BRAF}	NA
EVT-P-0024995-T01-IM6-121	P-0024995-T01-IM6		CMAS-BRAF	CMAS	unknown	three_prime	140488046	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		BRAF (NM_004333) rearrangement: t(7;12)(q34;p12.1)(chr7:g.140488046::chr12:g.22218755) Transcript Fusion {CMAS:BRAF}	NA
EVT-P-0039648-T01-IM6-122	P-0039648-T01-IM6		CUL1-BRAF	CUL1	in-frame	three_prime	140492893	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950:CUL1_c.1140+1215: BRAFInv Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-0039648-T03-IM6-123	P-0039648-T03-IM6		CUL1-BRAF	CUL1	in-frame	three_prime	140492893	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950:CUL1_c.1140+1215: BRAFInv Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-0064088-T01-IM7-124	P-0064088-T01-IM7		CUL1-BRAF	CUL1	in-frame	three_prime	140489451	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+1956:CUL1_c.1141-2067: BRAFInv Protein Fusion: in frame {CUL1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006 8173-T01-I M7-125	P-0068 173-T0 1-IM7		CUL1-BRAF	CUL1	in-frame	three_prime	140491990	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+191:CUL1_c.1140+2118:BRAFInv Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-005 0196-T01-I M6-126	P-0050 196-T0 1-IM6		DBI-BRAF	DBI	in-frame	three_prime	140486284	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		DBI (NM_001178017) - BRAF (NM_004333) fusion: t(2;7)(q14.2;q34)(chr2:g.120128894::chr7:g.140486284) Protein Fusion: in frame {DBI:BRAF}	NA
EVT-P-006 4588-T01-I M7-127	P-0064 588-T0 1-IM7		DOCK4-BRAF	DOCK4	out-of-frame	three_prime	140483872	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		DOCK4 (NM_014705) - BRAF (NM_004333) fusion: c.4650+298:DOCK4_c.1178-915:BRAFdup Protein Fusion: out of frame {DOCK4:BRAF}	NA
EVT-P-001 4833-T02-I M6-128	P-0014 833-T0 2-IM6		ERC1-BRAF	ERC1	in-frame	three_prime	140490668	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ERC1 (NM_178040) - BRAF (NM_004333) fusion: t(7;12)(q34;p13.33)(chr7:g.140490668::chr12:g.1246645) Protein Fusion: in frame {ERC1:BRAF}	NA
EVT-P-003 6937-T01-I M6-129	P-0036 937-T0 1-IM6		EXOC4-BRAF	EXOC4	out-of-frame	three_prime	140490241	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EXOC4 (NM_021807) - BRAF (NM_004333) fusion: c.1514+93469:EXOC4_c.1141-2857:BRAFInv Protein Fusion: out of frame {EXOC4:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 4206-T01-IM6-130	P-0024 206-T01-IM6		EXOC4-BRAF	EXOC4	out-of-frame	three_prime	140487956	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EXOC4 (NM_021807) - BRAF (NM_004333) rearrangement: c.87-8072:EXOC4_c.1141-572:BRAFInv Protein Fusion: out of frame {EXOC4:BRAF}	NA
EVT-P-004 7918-T04-IM7-131	P-0047 918-T04-IM7		FAM131B-BRAF	FAM131B	in-frame	three_prime	140487437	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_001031690) - BRAF (NM_004333) fusion: c.175-20:FAM131B_c.1141-53:BRAFdel Protein Fusion: in frame {FAM131B:BRAF}	NA
EVT-P-004 7918-T02-IM6-132	P-0047 918-T02-IM6		FAM131B-BRAF	FAM131B	in-frame	three_prime	140487437	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_001031690) - BRAF (NM_004333) fusion: c.175-20:FAM131B_c.1141-53:BRAFdel Protein Fusion: in frame {FAM131B:BRAF}	NA
EVT-P-004 4845-T01-IM6-134	P-0044 845-T01-IM6		GTF2I-BRAF	GTF2I	unknown	three_prime	140485631	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GTF2I (NM_032999) - BRAF (NM_004333) fusion: c.768:GTF2I_c.1177+1717:BRAFInv Protein Fusion: mid-exon {GTF2I:BRAF}	NA
EVT-P-000 0828-T05-IM6-135	P-0000 828-T05-IM6		GTF2IRD1-BRAF	GTF2IRD1	in-frame	three_prime	140483546	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GTF2IRD1 (NM_005685) - BRAF (NM_004333) rearrangement: c.2320+14147:GTF2IRD1_c.1178-589:BRAFInv Protein Fusion: in frame {GTF2IRD1:BRAF}	NA

event_id	sampl_e_id	pa_tie_nt_i_d	fusion_nam_e	part_ner_ge_ne	fra_me_st_atu_s	tar_get_ro_le	breakp_o_int_ge_nomic	break_p_o_int_exo_n	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-004 6745-T01-I M6-136	P-0046 745-T0 1-IM6		KCND 2-BRA F	KCN D2	out-of-fr ame	thr_ee_pr_im_e	140485 867	9	393	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		KCND2 (NM_012281) - BRAF (NM_004333) fusion: c.1115+91797:KCND2_c.11 77+1481:BRACFInv Protein Fusion: out of frame {KCND2:BRACF}	NA
EVT-P-005 6952-T01-I M6-138	P-0056 952-T0 1-IM6		KIAA1 549-B RAF	KIA A15 49	in-fr ame	thr_ee_pr_im_e	140493 821	8	380	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-141:KIAA154 9_c.1140+287:BRACFdup Protein Fusion: in frame {KIAA1549:BRACF}	NA
EVT-P-004 4270-T02-I M6-139	P-0044 270-T0 2-IM6		KIAA1 549-B RAF	KIA A15 49	in-fr ame	thr_ee_pr_im_e	140491 645	8	380	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1560:KIAA1 549_c.1140+2463:BRACFdu p Protein Fusion: in frame {KIAA1549:BRACF}	NA
EVT-P-002 7592-T01-I M6-140	P-0027 592-T0 1-IM6		KIAA1 549-B RAF	KIA A15 49	in-fr ame	thr_ee_pr_im_e	140492 343	8	380	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5247+769:KIAA15 49_c.1140+1765:BRACFdup Protein Fusion: in frame {KIAA1549:BRACF}	NA
EVT-P-005 5099-T03-I M6-141	P-0055 099-T0 3-IM6		KIAA1 549-B RAF	KIA A15 49	in-fr ame	thr_ee_pr_im_e	140490 663	9	381	True	retai ned	1.0	False	Protein kinase domai n	RAS -bin ding dom ain; Cyst eine -rich dom ain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+1357:KIAA1 549_c.1141-3279:BRACFdup Protein Fusion: in frame {KIAA1549:BRACF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-005 8584-T01-I M7-142	P-0058 584-T0 1-IM7		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140491493	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1663:KIAA1549_c.1140+2615:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-006 6030-T01-I M7-143	P-0066 030-T0 1-IM7		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140482785	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1172:KIAA1549_c.1314+36:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-003 5153-T01-I M6-144	P-0035 153-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140491093	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon15 fused with BRAF exon9) : c.4930-2046:KIAA1549_c.1140+3015:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-003 5153-T03-I M6-145	P-0035 153-T0 3-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140491093	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2046:KIAA1549_c.1140+3015:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-004 9925-T01-I M6-146	P-0049 925-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140489497	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+301:KIAA1549_c.1141-2113:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 5844-T03-I M7-147	P-0035 844-T0 3-IM7		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140493285	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140+823:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-003 5844-T04-I M7-148	P-0035 844-T0 4-IM7		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140493285	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140+823:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-001 7693-T02-I M6-149	P-0017 693-T0 2-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140489537	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-495:KIAA1549_c.1141-2153:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-006 2963-T01-I M7-150	P-0062 963-T0 1-IM7		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140482198	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-27:KIAA1549_c.1314+623:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-001 2575-T01-I M5-151	P-0012 575-T0 1-IM5		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140490295	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+793:KIAA1549_c.1141-2911:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-005 2411-T05-I M7-152	P-0052 411-T0 5-IM7		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140493530	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140+578:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-003 4117-T01-I M6-153	P-0034 117-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140483812	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4346-57:KIAA1549_c.1178-855:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-005 2411-T03-I M6-154	P-0052 411-T0 3-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140493530	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140+578:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-000 9097-T01-I M5-155	P-0009 097-T0 1-IM5		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140488817	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.5247+3837:KIAA1549_c.1141-1433:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-000 5121-T02-I M5-156	P-0005 121-T0 2-IM5		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140491248	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+2454:KIAA1549_c.1140+2860:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006 4872-T01-I M7-157	P-0064 872-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489277	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4032+4347:KIAA1549_c.1141-1893:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-001 3777-T01-I M5-158	P-0013 777-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140481674	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549(NM_001164665) - BRAF (NM_004333) rearrangement: c.5248-2734:KIAA1549_c.1315-181:BRADFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-004 0760-T01-I M6-159	P-0040 760-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140481540	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2627:KIAA1549_c.1315-47:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-001 4191-T01-I M6-160	P-0014 191-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492679	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion : c.5248-320: KIAA1549_c.1140+1429:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-001 4381-T01-I M6-161	P-0014 381-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492888	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused to BRAF exon 9 -18): c.4930-2082:KIAA1549_c.1140+1220:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 8611-T01-IM6-162	P-0028 611-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482786	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-001 5540-T01-IM6-163	P-0015 540-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489119	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.4930-69:KIAA1549_c.1141-1735:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-006 2070-T01-IM7-164	P-0062 070-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490443	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1064:KIAA1549_c.1141-3059:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-002 2404-T01-IM6-165	P-0022 404-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482177	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 11-18) : c.5248-2270:KIAA1549_c.1314+644:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-000 4032-T01-IM5-166	P-0004 032-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140487892	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon 16 fused to BRAF exon 9) : c.5248-1949:KIAA1549_c.1141-508:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 3372-T01-I M6-167	P-0043 372-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140487523	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2204:KIAA1549_c.1141-139:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-002 8611-T02-I M6-168	P-0028 611-T0 2-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140482786	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-004 3874-T01-I M6-169	P-0043 874-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140490001	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-2907:KIAA1549_c.1141-2617:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-006 9024-T04-I M7-170	P-0069 024-T0 4-IM7		KIAA1549-B RAF	KIAA1549	unknown	three_prime	140492117	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5059:KIAA1549_c.1140+1991:BRAFdup Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-002 9361-T01-I M6-171	P-0029 361-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	three_prime	140489625	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-2667:KIAA1549_c.1141-2241:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 6557-T01-I M6-172	P-0016 557-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	threonine_prime	140488576	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in-frame to BRAF exons 9-18) : c.4929+2955: KIAA1549_c.1141-1192:BR AFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-003 2982-T01-I M6-173	P-0032 982-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	threonine_prime	140491506	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5248-10:KIAA1549_c.1140+2602:BR AFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-006 4310-T04-I M7-174	P-0064 310-T0 4-IM7		KIAA1549-B RAF	KIAA1549	in-frame	threonine_prime	140493446	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+3560:KIAA1549_c.1140+662:BR AFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-002 1127-T01-I M6-175	P-0021 127-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	threonine_prime	140491741	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-10 fused with BRAF exons 9-18) : c.4032+3996:KIAA1549_c.1140+2367:BR AFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-001 7693-T01-I M6-176	P-0017 693-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	threonine_prime	140489537	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (c.1141-2153): c.4930-495: KIAA1549_c.1141-2153:BR AFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 0125-T01-I M6-177	P-0020 125-T0 1-IM6		KIAA1549-B RAF	KIAA1549	unknown	threepri	140492175	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-17 fused to BRAF exons 9-18): c.5260:KIAA1549_c.1140+1933:BRAFdup Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-004 5683-T02-I M6-178	P-0045 683-T0 2-IM6		KIAA1549-B RAF	KIAA1549	unknown	threepri	140492850	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5189:KIAA1549_c.1140+1258:BRAFdup Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-001 9127-T01-I M6-179	P-0019 127-T0 1-IM6		KIAA1549-B RAF	KIAA1549	in-frame	threepri	140490657	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 9-18): c.5247+2110:KIAA1549_c.1141-3273:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-006 6022-T01-I M7-180	P-0066 022-T0 1-IM7		KIAA1549-B RAF	KIAA1549	in-frame	threepri	140488035	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-4074:KIAA1549_c.1141-651:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-003 6680-T01-I M6-181	P-0036 680-T0 1-IM6		KLHL12-BR AF	KLHL12	in-frame	threepri	140493984	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KLHL12 (NM_021633) - BRAF (NM_004333) fusion (KLHL12 exons 1-4 fused to BRAF exons 9-18): t(1;7)(q32.1;q34)(chr1:g.202880970:chr7:g.140493984) Protein Fusion: in frame {KLHL12:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0047833-T02-IM6-183	P-0047833-T02-IM6		LRBA-BRAF	LRBA	in-frame	three_prime	140493444	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		LRBA (NM_001199282) - BRAF (NM_004333) fusion: t(4;7)(q31.3;q34)(chr4:g.151502269::chr7:g.140493444) Protein Fusion: in frame {LRBA:BRAF}	NA
EVT-P-0006412-T01-IM5-184	P-0006412-T01-IM5		LUC7L2-BRAF	LUC7L2	unknown	five_prime	140501109	6	287	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		LUC7L2 (NM_001244584) - BRAF (NM_004333) Rearrangement : c.1008-2090:LUC7L2_c.860+103:BR AFdel Antisense fusion	NA
EVT-P-0026158-T04-IM6-185	P-0026158-T04-IM6		MAD1L1-BRAF	MAD1L1	in-frame	three_prime	140491749	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MAD1L1 (NM_001013837) - BRAF (NM_004333) fusion: c.1417-9863:MAD1L1_c.1140+2359:BRAFdup Protein Fusion: in frame {MAD1L1:BRAF}	NA
EVT-P-0026158-T01-IM6-186	P-0026158-T01-IM6		MAD1L1-BRAF	MAD1L1	in-frame	three_prime	140491749	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MAD1L1 (NM_001013837) - BRAF (NM_004333) Rearrangement : c.1417-9863_c.1140+2359dup Protein Fusion: in frame {MAD1L1:BRAF}	NA
EVT-P-0006601-T03-IM6-187	P-0006601-T03-IM6		MINDY4-BRAF	MINDY4	in-frame	three_prime	140490696	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM188B (NM_032222) - BRAF(NM_004333) Fusion (FAM188B exon5 fused with BRAF exon9) : c.664-366:FAM188B_c.1141-3312:BR AF Protein Fusion: in frame {FAM188B:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 3220-T01-I M6-188	P-0023 220-T0 1-IM6		MKLN 1-BRAF	MKLN1	in-frame	three_prime	140496796	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKLN1 (NM_013255) - BRAF (NM_004333) rearrangement: c.98+4070:MKLN1_c.981-2529:BRAFinv Protein Fusion: in frame {MKLN1:BRAF}	NA
EVT-P-003 5393-T02-I M6-189	P-0035 393-T0 2-IM6		MKRN 1-BRAF	MKRN1	in-frame	three_prime	140488436	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.771+833:MKRN1_c.1141-1052:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-000 9024-T01-I M5-190	P-0009 024-T0 1-IM5		MKRN 1-BRAF	MKRN1	in-frame	three_prime	140493485	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.772-184G:MKRN1_c.1140+623:BRAFdup Protein fusion: in frame (MKRN1-BRAF)	NA
EVT-P-000 8083-T03-I M6-191	P-0008 083-T0 3-IM6		MKRN 1-BRAF	MKRN1	out-of-frame	three_prime	140484511	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.315-432:MKRN1_c.1178-1554:BRAFdup Protein Fusion: out of frame {MKRN1:BRAF}	NA
EVT-P-000 1036-T01-I M3-192	P-0001 036-T0 1-IM3		MKRN 1-BRAF	MKRN1	in-frame	three_prime	140482250	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA Protein fusion: in frame (MKRN1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 7135-T01-I M6-193	P-0047 135-T0 1-IM6		MKRN 1-BRAF	MKRN1	in-frame	three_prime	140492 510	8	380	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-972:MKRN1_c.1140+ 1598:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-000 4373-T02-I M5-194	P-0004 373-T0 2-IM5		MKRN 1-BRAF	MKRN1	unknown	three_prime	140434 466	18	744	False	lost	0.0	False		Protein kinase domain; RAS- binding domain; Cysteine- rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.314+4351 :MKRN1_c.2232:BRAFdup Protein fusion: mid-exon (MKRN1-BRAF)	NA
EVT-P-000 9024-T02-I M6-195	P-0009 024-T0 2-IM6		MKRN 1-BRAF	MKRN1	in-frame	three_prime	140493 485	8	380	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-185:MKRN1_c.1140+ 623:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-003 1896-T02-I M6-196	P-0031 896-T0 2-IM6		MKRN 1-BRAF	MKRN1	in-frame	three_prime	140482 581	10	438	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion (MKRN1 exons 1-4 fused to BRAF exons 11-18): c.772- 1018:MKRN1_c.1314+240: BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-000 8083-T02-I M6-197	P-0008 083-T0 2-IM6		MKRN 1-BRAF	MKRN1	out-of-frame	three_prime	140484 505	10	393	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion (MKRN1 exons 1-2 fused with BRAF exons 10-18): c. 315-428:MKRN1_c.1178-15 48:BRAFdup Protein Fusion: out of frame {MKRN1:BRAF}	NA
EVT-P-001 3536-T01-I M5-198	P-0013 536-T0 1-IM5		MKRN 1-BRAF	MKRN1	in-frame	three_prime	140483 618	10	393	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.544+172: MKRN1_c.1178-661:BRAFdup Protein fusion: in frame (MKRN1-BRAF)	NA
EVT-P-004 8036-T01-I M6-199	P-0048 036-T0 1-IM6		NEO1 -BRAF	NEO1	in-frame	three_prime	140497 651	7	327	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		NEO1 (NM_002499) - BRAF (NM_004333) fusion: t(7;15)(q34;q24.1)(chr7:g.14 0497651::chr15:g.73575682) Protein Fusion: in frame {NEO1:BRAF}	NA
EVT-P-004 3972-T03-I M6-200	P-0043 972-T0 3-IM6		NRF1- BRAF	NRF1	in-frame	three_prime	140487 712	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		NRF1 (NM_005011) - BRAF (NM_004333) fusion: c.607- 7410:NRF1_c.1141-328:BR AFinv Protein Fusion: in frame {NRF1:BRAF}	NA
EVT-P-006 5446-T01-I M7-201	P-0065 446-T0 1-IM7		NRF1- BRAF	NRF1	in-frame	three_prime	140483 810	10	393	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		NRF1 (NM_005011) - BRAF (NM_004333) fusion: c.1349 -10893:NRF1_c.1178-853:B RAFinv Protein Fusion: in frame {NRF1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 9002-T01-I M7-202	P-0059 002-T0 1-IM7		OSBP L7-BRAF	OSB PL7	in-frame	three_prime	140491 530	8	380	True	retained	1.0	False	Protein kinase domain	RAS -binding domain; Cysteine -rich domain		OSBPL7 (NM_145798) - BRAF (NM_004333) fusion: t(7;17)(q34;q21.32)(chr7:g.1 40491530::chr17:g.4589038 1) Protein Fusion: in frame {OSBPL7:BRAF}	NA
EVT-P-005 4958-T02-I M6-203	P-0054 958-T0 2-IM6		PAK2- BRAF	PAK 2	in-frame	three_prime	140486 065	9	393	True	retained	1.0	False	Protein kinase domain	RAS -binding domain; Cysteine -rich domain		PAK2 (NM_002577) - BRAF (NM_004333) rearrangement: t(3;7)(q29;q 34)(chr3:g.196530579::chr7: g.140486065) Protein Fusion: in frame {PAK2:BRAF}	NA
EVT-P-000 8789-T01-I M5-205	P-0008 789-T0 1-IM5		PHTF 2-BRAF	PHT F2	in-frame	three_prime	140489 236	9	381	True	retained	1.0	False	Protein kinase domain	RAS -binding domain; Cysteine -rich domain		PHTF2 (NM_001127358) - BRAF (NM_004333) rearrangement: c.2224-367: PHTF2_c.1141-1852:BRAF Protein fusion: in frame {PHTF2-BRAF}	NA
EVT-P-004 4819-T01-I M6-206	P-0044 819-T0 1-IM6		PLPP 1-BRAF	PLP P1	out-of-frame	three_prime	140482 123	11	439	True	retained	1.0	False	Protein kinase domain	RAS -binding domain; Cysteine -rich domain		PPAP2A (NM_176895) - BRAF (NM_004333) fusion: t(5;7)(q11.2;q34)(chr5:g.547 89192::chr7:g.140482123) Protein Fusion: out of frame {PPAP2A:BRAF}	NA
EVT-P-000 9694-T01-I M5-207	P-0009 694-T0 1-IM5		PRKAR1B- BRAF	PRK AR1 B	in-frame	three_prime	140491 855	8	380	True	retained	1.0	False	Protein kinase domain	RAS -binding domain; Cysteine -rich domain		PRKAR1B (NM_001164758) -BRAF (NM_004333) Rearrangement : c.892-255 1:PRKAR1B_c.1140+2253: BRAFdup Protein fusion: in frame {PRKAR1B-BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000 8567-T01-I M5-208	P-0008 567-T01-IM5		PRKAR2B-BRAF	PRKAR2B	in-frame	three_prime	140486416	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRKAR2B (NM_002736) - BRAF (NM_004333) rearrangement: c.307+8063:PRKAR2B_c.1177+932:BRAFInv Protein fusion: in frame (PRKAR2B-BRAF)	NA
EVT-P-004 1181-T01-I M6-209	P-0041 181-T01-IM6		PTPRZ1-BRAF	PTPRZ1	in-frame	three_prime	140492428	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PTPRZ1 (NM_002851) - BRAF (NM_004333) fusion: c.5367+457:PTPRZ1_c.1140+1680:BRAFInv Protein Fusion: in frame {PTPRZ1:BRAF}	NA
EVT-P-001 2884-T01-I M5-210	P-0012 884-T01-IM5		PWWP2A-BRAF	PWWP2A	in-frame	three_prime	140495243	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PWWP2A (NM_001130864) - BRAF (NM_004333) Rearrangement : t(5;7)(q33.3;q34)(chr5:g.159539324::chr7:g.140495243) Protein fusion: in frame (PWWP2A-BRAF)	NA
EVT-P-006 8260-T01-I M7-211	P-0068 260-T01-IM7		RBMS3-BRAF	RBMS3	in-frame	three_prime	140482707	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RBMS3 (NM_001003793) - BRAF (NM_004333) fusion: t(3;7)(p24.1;q34)(chr3:g.30030473::chr7:g.140482707) Protein Fusion: in frame {RBMS3:BRAF}	NA
EVT-P-006 4764-T01-I M7-212	P-0064 764-T01-IM7		RGS3-BRAF	RGS3	out-of-frame	three_prime	140493282	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RGS3 (NM_144488) - BRAF (NM_004333) fusion: t(7;9)(q34;q32)(chr7:g.140493282::chr9:g.116291333) Protein Fusion: out of frame {RGS3:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-004 1308-T01-I M6-213	P-0041 308-T0 2-IM6		RRBP1-BRAF	RRBP1	in-frame	three_prime	140485917	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RRBP1 (NM_004587) - BRAF (NM_004333) fusion: t(7;20)(q34;p12.1)(chr7:g.140485917::chr20:g.17634546) Protein Fusion: in frame {RRBP1:BRAF}	NA
EVT-P-000 7392-T01-I M5-214	P-0007 392-T0 1-IM5		SCRIB-BRAF	SCRIB	in-frame	three_prime	140482150	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SCRIB (NM_182706) - BRAF (NM_004333) rearrangement: t(7;8)(q34;q24.3)(chr7:g.140482150::chr8:g.144876675) Protein fusion: in frame (SCRIB-BRAF)	NA
EVT-P-005 6639-T01-I M6-215	P-0056 639-T0 1-IM6		SETBP1-BRAF	SETBP1	in-frame	three_prime	140486594	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SETBP1 (NM_015559) - BRAF (NM_004333) fusion: t(7;18)(q34;q12.3)(chr7:g.140486594::chr18:g.42638299) Protein Fusion: in frame {SETBP1:BRAF}	NA
EVT-P-004 8950-T01-I M6-216	P-0048 950-T0 1-IM6		SND1-BRAF	SND1	in-frame	three_prime	140492033	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+30037:SND1_c.1140+2075:BRAFinv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-001 2395-T01-I M5-217	P-0012 395-T0 1-IM5		SND1-BRAF	SND1	in-frame	three_prime	140493527	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1152+27174:SND1_c.1140+581:BRAFinv Protein fusion: in frame (SND1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-000 9619-T01-I M5-218	P-0009 619-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140482494	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1968+2394:SND1_c.1314+327:BRAFinv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-005 2299-T04-I M6-219	P-0052 299-T04-IM6		SND1-BRAF	SND1	unknown	three_prime	140482935	10	400	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1038+2929:SND1_c.1200:BRAFinv Protein Fusion: mid-exon {SND1:BRAF}	NA
EVT-P-000 1687-T01-I M3-220	P-0001 687-T01-IM3		SND1-BRAF	SND1	in-frame	three_prime	140490224	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) inversion: c.1152+28381_c.1141-2840inv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-000 4103-T01-I M5-221	P-0004 103-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140481685	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1038+5799:SND1_c.1315-192:BRAFinv. Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-002 9727-T01-I M6-222	P-0029 727-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140482257	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+12095:SND1_c.1314+564:BRAFinv Protein Fusion: in frame {SND1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-006 8194-T01-I M7-223	P-0068 194-T0 1-IM7		SND1- BRAF	SND 1	in-frame	three_prime	140489 175	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+17686:SND1_c.1141-1791:BRAFInv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-000 8140-T01-I M5-224	P-0008 140-T0 1-IM5		SND1- BRAF	SND 1	in-frame	three_prime	140489 439	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-9 fused with BRAF exons 9-18) : c.1039-3849:SND1_c.1141-2055:BRAFInv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-003 4168-T01-I M6-225	P-0034 168-T0 1-IM6		SND1- BRAF	SND 1	in-frame	three_prime	140489 421	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused to BRAF exons 9-18): c.1152+6972:SND1_c.1141-2037:BRAFInv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-002 5075-T01-I M6-226	P-0025 075-T0 1-IM6		SND1- BRAF	SND 1	in-frame	three_prime	140490 546	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1- 9 with BRAF exons 9-18) : c.1039-2471:SND1_c.1141-3162:BRAFInv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-000 2071-T02-I M5-227	P-0002 071-T0 2-IM5		SND1- BRAF	SND 1	in-frame	three_prime	140489 292	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused with BRAF exons 9-18) : c.1152+18606:SND1_c.1141-1908:BRAFInv Protein fusion: in frame (SND1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-000 2197-T01-I M3-228	P-0002 197-T0 1-IM3		SND1- BRAF	SND 1	in-frame	five_prime	140488 303	9	381	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		c.1528-8436:SND1_c.1141-922:BRAFinv Protein fusion: in frame (BRAF-SND1)	NA
EVT-P-000 7533-T01-I M5-229	P-0007 533-T0 1-IM5		SND1- BRAF	SND 1	in-frame	three_prime	140482 226	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 with BRAF exons 11-18) : c.1152+31738:SND1_c.1314+595:BRAFinv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-003 6567-T01-I M6-230	P-0036 567-T0 1-IM6		SND1- BRAF	SND 1	in-frame	three_prime	140493 552	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) Fusion (SND1 exon10 fused with BRAF exon9) : c.1153-22013:SND1_c.1140+556:BRAFinv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-003 4845-T01-I M6-231	P-0034 845-T0 1-IM6		SND1- BRAF	SND 1	in-frame	three_prime	140489 241	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1527+3658:SND1_c.1141-1857:BRAFinv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-005 4364-T01-I M6-232	P-0054 364-T0 1-IM6		SNX8- BRAF	SNX 8	in-frame	three_prime	140488 730	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SNX8 (NM_013321) - BRAF (NM_004333) fusion: c.1284+565:SNX8_c.1141-1346:BRAFdup Protein Fusion: in frame {SNX8:BRAF}	NA
EVT-P-002 6773-T01-I M6-234	P-0026 773-T0 1-IM6		SVOP L-BRAF	SVO PL	unknown	five_prime	140482 097	11	439	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) rearrangement: c.1315-604: BRAF_chr7:g.138277426del Transcript Fusion {BRAF:SVOPL}	NA

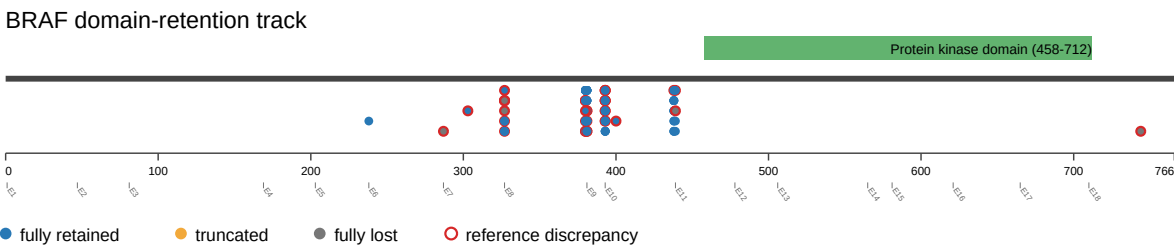
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-005 1214-T01-I M6-235	P-0051 214-T0 1-IM6		TMPR SS2-B RAF	TMP RSS 2	in-frame	three_prime	140502 592	6	238	True	retained	1.0	False	Protein kinase domain	RAS- binding domain	Cysteine-rich domain	TMPRSS2 (NM_001135099) - BRAF (NM_004333) fusion: t(7;21) (q34;q22.3)(chr7:g.1405025 92::chr21:g.42867846) Protein Fusion: in frame {TMPRSS2:BRAF}	NA
EVT-P-001 8524-T01-I M6-236	P-0018 524-T0 1-IM6		TMPR SS2-B RAF	TMP RSS 2	out-of-frame	three_prime	140490 337	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		TMPRSS2 (NM_001135099) - BRAF (NM_004333) Rearrangement : t(21;7)(q2 2.3;q34)(chr21:g.42873801: :chr7:g.140490337) Protein Fusion: out of frame {TMPRSS2:BRAF}	NA
EVT-P-002 2778-T01-I M6-237	P-0022 778-T0 1-IM6		TMPR SS2-B RAF	TMP RSS 2	in-frame	three_prime	140483 147	10	393	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		TMPRSS2 (NM_001135099) - BRAF (NM_004333) Rearrangement : t(7;21)(q3 4;q22.3)(chr7:g.140483147: :chr21:g.42878796) Protein Fusion: in frame {TMPRSS2:BRAF}	NA
EVT-P-004 3836-T01-I M6-239	P-0043 836-T0 1-IM6		TRA2 B-BR AF	TRA 2B	in-frame	three_prime	140491 978	8	380	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		TRA2B (NM_004593) - BRAF (NM_004333) fusion: t(3;7)(q27.2;q34)(chr3:g.185 650761::chr7:g.140491978) Protein Fusion: in frame {TRA2B:BRAF}	NA
EVT-P-005 4430-T02-I M6-240	P-0054 430-T0 2-IM6		TRIM2 4-BR AF	TRI M24	in-frame	three_prime	140487 875	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2719-489:TRIM24_c.1141 -491:BRAFinv Protein Fusion: in frame {TRIM24:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 5699-T02-I M6-241	P-0025 699-T0 2-IM6		TRIM2 4-BRAF	TRIM24	in-frame	three_prime	140486 854	9	393	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2014+674:TRIM24_c.117 7+494:BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-004 5111-T01-I M6-242	P-0045 111-T0 1-IM6		TRIM2 4-BRAF	TRIM24	in-frame	three_prime	140488 049	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2257-760:TRIM24_c.1141 -665:BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-006 6835-T01-I M7-243	P-0066 835-T0 1-IM7		TRIM2 4-BRAF	TRIM24	in-frame	three_prime	140490 589	9	381	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.1879-502:TRIM24_c.1141 -3205:BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-006 8451-T01-I M7-244	P-0068 451-T0 1-IM7		TRIM2 4-BRAF	TRIM24	in-frame	three_prime	140482 765	10	438	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.484-860:TRIM24_c.1314+ 56:BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-003 0988-T01-I M6-245	P-0030 988-T0 1-IM6		TRIM2 4-BRAF	TRIM24	unknown	three_prime	140491 002	8	380	True	retained	1.0	False	Protein kinase domain	RAS- binding domain; Cysteine- rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2349:TRIM24_c.1140+31 06:BRAFInv Protein Fusion: mid-exon {TRIM24:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 2569-T01-I M6-246	P-0032 569-T0 1-IM6		TRIM24-BRAF	TRIM24	in-frame	three_prime	140487936	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) Rearrangement : c.1878+791:TRIM24_c.1141-552inv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-003 2569-T02-I M6-247	P-0032 569-T0 2-IM6		TRIM24-BRAF	TRIM24	in-frame	three_prime	140487936	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.1878+791:TRIM24_c.1141-552:BRAFinv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-001 9039-T01-I M6-248	P-0019 039-T0 1-IM6		UBE2H-BRAF	UBE2H	out-of-frame	three_prime	140499248	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		BRAF (NM_004333) - UBE2H (NM_003344) rearrangement: c.980+914: BRAF_c.299-4611:UBE2Hdup Protein Fusion: out of frame {UBE2H:BRAF}	NA
EVT-P-002 7762-T01-I M6-250	P-0027 762-T0 1-IM6		ZC3HAV1-BRAF	ZC3HAV1	in-frame	three_prime	140485946	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ZC3HAV1 (NM_020119) - BRAF (NM_004333) rearrangement: c.698-431:ZC3HAV1_c.1177+1402:BRAFdup Protein Fusion: in frame {ZC3HAV1:BRAF}	NA
EVT-P-000 8920-T01-I M5-251	P-0008 920-T0 1-IM5		ZNF207-BRAF	ZNF207	in-frame	three_prime	140485847	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ZNF207 (NM_001098507) - BRAF (NM_004333) rearrangement: t(7;17)(q34;q11.2)(chr7:g.140485847::chr17:g.30687533) Protein fusion: in frame (ZNF207-BRAF)	NA

Figures

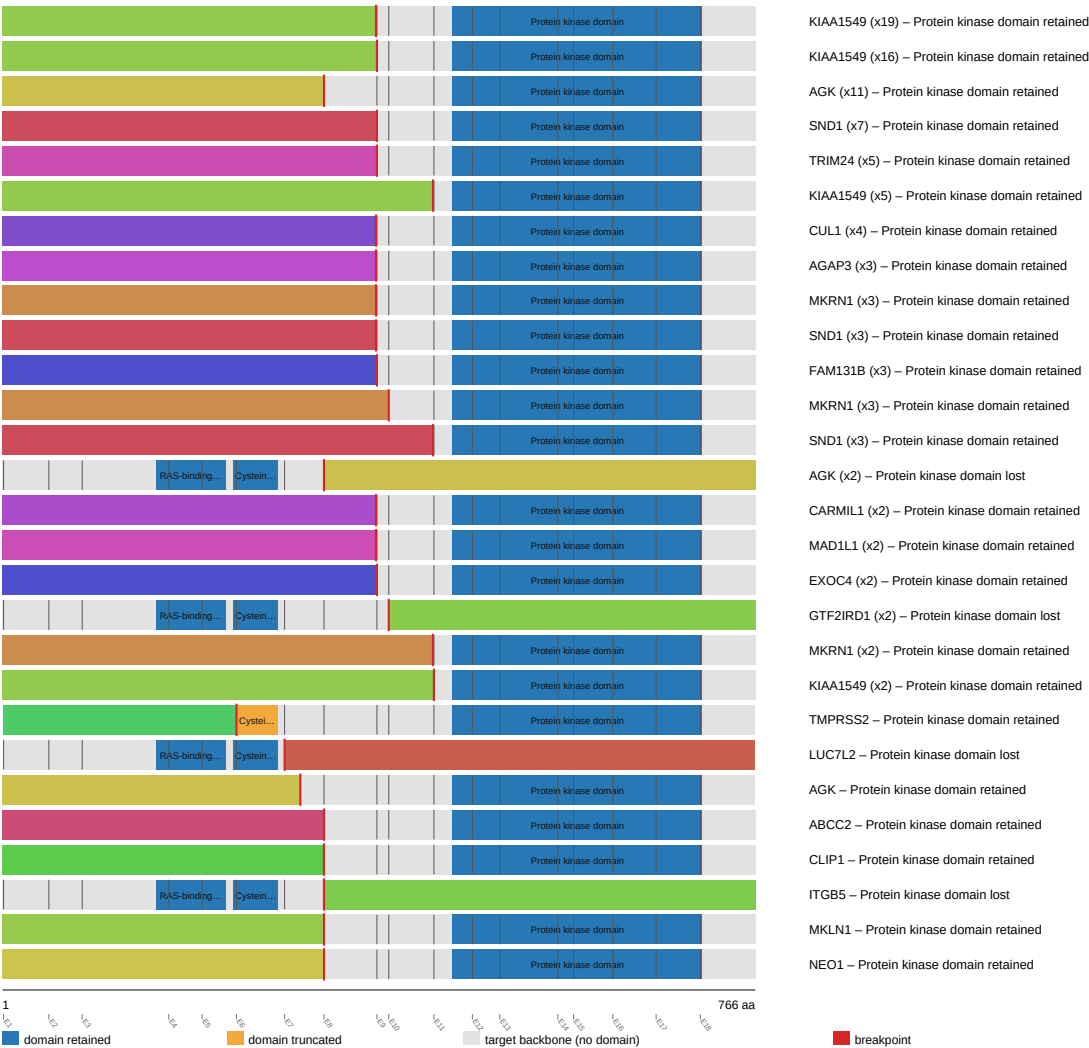
domain retention outliers



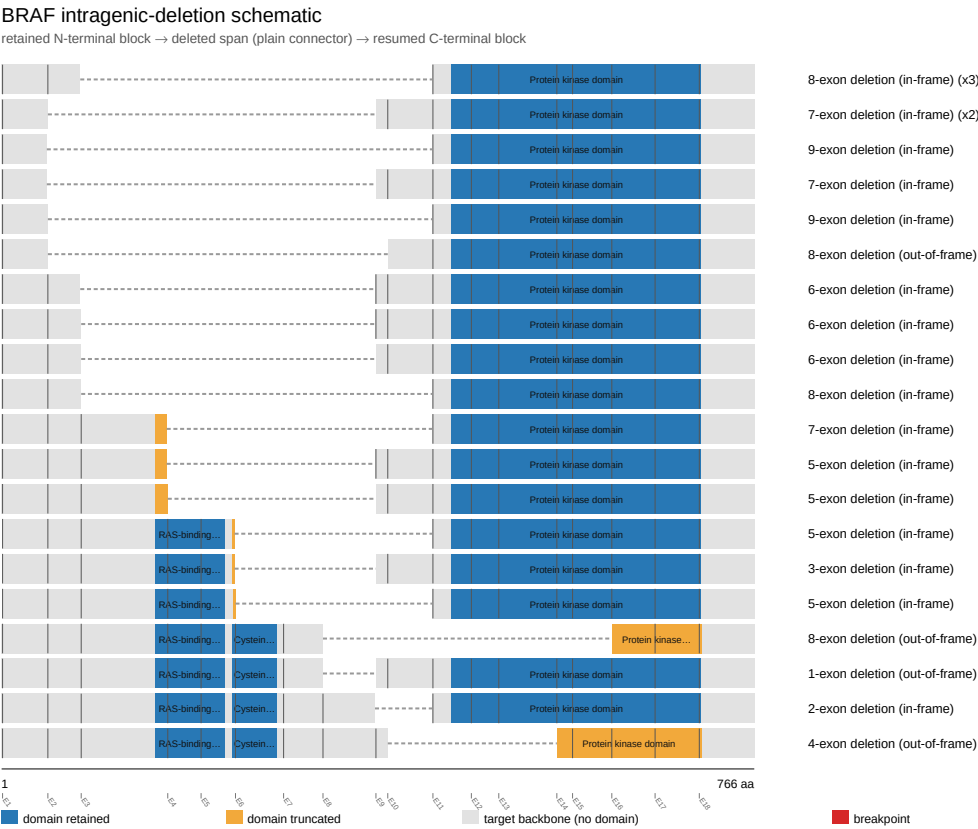
fusion schematic

BRAF fusion-transcript schematic

partner block → breakpoint → retained BRAF portion (domain-colored, exon ticks)



intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

